ENGG 112 Assignment

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Q no.1 →

a) Resistance of a lamp $(R) = 28\frac{1}{8}\,\Omega = \frac{225}{8}\,\Omega$

Total resistances in series $= \frac{225}{8} \times 8 = 225\,\Omega$

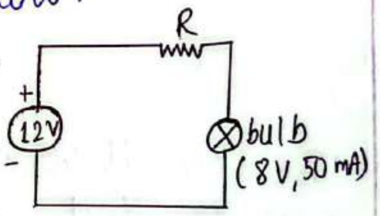
$\therefore$ Current through the bulbs $(I) = \frac{120}{225} = 0.533\,A$

b) Power in each bulb $= I^2R = \left(\frac{8}{15}\right)^2 \times \frac{225}{8} = 8\,W$

c) voltage drop across each bulb $= I.R = \frac{8}{15} \times \frac{225}{8} = 15\,V$

d) As all the bulbs are in series, if one bulb burns out the circuit will be open and none of the bulb would glow.

Q no.2 → The voltage divider circuit for the given condition is as shown below.




As the bulb requires 8V, the resistor should have a voltage drop $= (12-8)\,V = 4\,V$

Also as the resistor are in series with the bulb same current flow in both resistor and bulb i.e. 50mA.

$\therefore$ The resistance, $R = \frac{4V}{50mA} = 80\,\Omega$

Also the power in the resistance $= 4V \times 50\,mA$
$= 0.2\,W$

$\therefore$ From the available resistors, the minimum wattage rating of the resistor R will be 0.25 W.

Q no. 3

(a) The current through each 60 W lamp $= \frac{60W}{120V} = 0.5A$

The current through 400W washer $= \frac{400 W}{120 V} = 3.33A$

The current through 360W TV $= \frac{360 W}{120 V} = 3 A$

(b) The total current drawn from the 120 V source

$= 10 \times 0.5 + 3.33 + 3$

$= 11.33 A$

The circuit breaker is rated as 20A i.e. it will open when the current exceeds 20A. For the given load the maximum current $= 11.33 A < 20 A$ i.e. less than the circuit breaker rating so the circuit breaker will not open.

(c) The total resistance of the network $= \frac{\text{Total Voltage}}{\text{Total current}}$

$= \frac{120}{11.33} = 10.588 \Omega$

d) The circuit connection & the circuit breaker rating specifies that, the maximum power that can be supplied by the source $= 120 V \times 20A = 2400 W$

The total power of the load $= 60 W \times 10 + 400 W + 360 W$

$= 1360 W$

$\therefore$ other loads upto $2400 - 1360 = 1040 W$ can also be connected to the source.

Q.no. 4

a) The open circuit voltage $V_L = \frac{4.7\times10^3\times9}{2.2\times10^3+4.7\times10^3} = 6.13\,V$

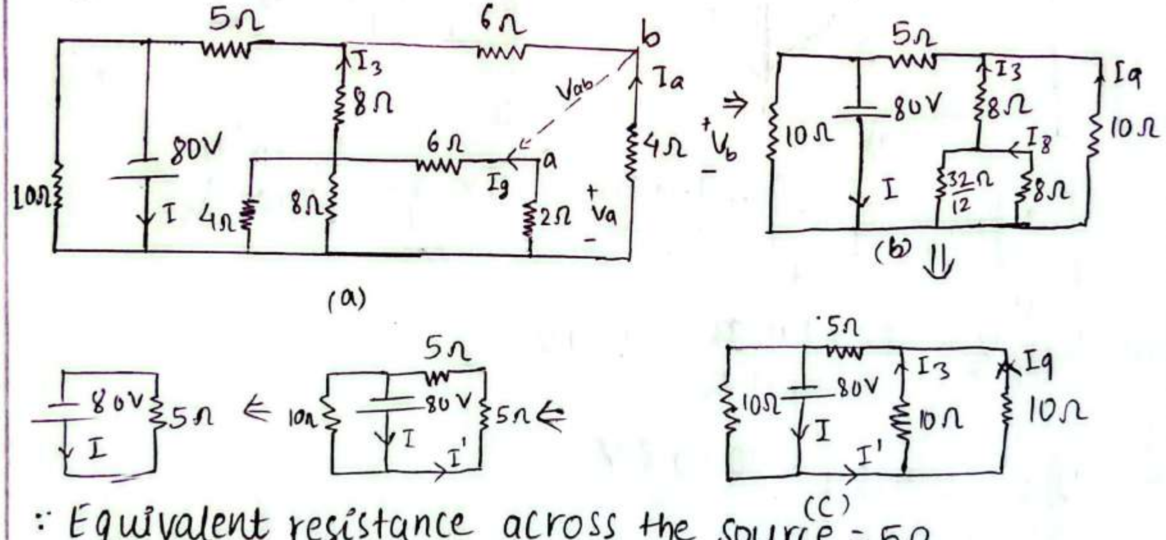
(b) If 2.2kΩ is short circuited, the voltage sources becomes parallel to 4.7 kΩ resistance,

$\therefore V_L = 9V$

(c) If 4.7 kΩ resistor is open circuited, there will be no current in the circuit i.e. no voltage drop in resistors, $\therefore V_L = 9V$

Q.no. 5

a) The current I can be determined by finding the total resistance in parallel to the voltage source. Thus to find the equivalent resistance the circuit is reduced as under:




$\therefore$ Equivalent resistance across the source = 5Ω

$\therefore$ The current $I = 80/5 = 16\ A$

(b) As per the figure (d) the current $I' = \frac{16\times10}{10+10} = 8A$

$\therefore$ The current $I_3$ as shown in figure (c) $= \frac{10.I'}{10+10} = \frac{10\times8}{20} = 4A$

Ily, current $I_q = \frac{10\times8}{20} = 4A$

[or, $I_q = I' - I_3 = 8 - 4 = 4A$

(c) As per the figure (b) the current $I_3$ divides in two lower branch and current of one lower branch is $I_8$.

$$\because I_8 = \frac{\frac{32}{12} \times I_3}{\frac{32}{12} + 8} = \frac{\frac{32}{12} \times 4}{\frac{32}{12} + 8} = 1A$$

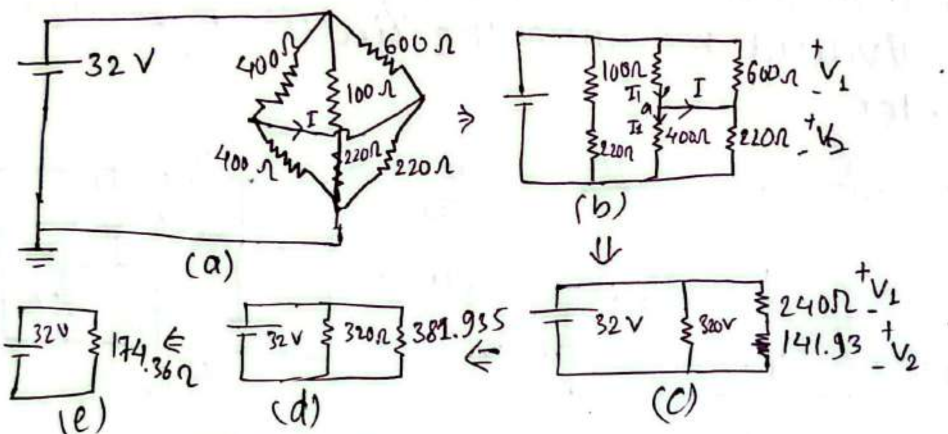
(d) The voltage $V_{ab}$ is the difference of voltage $V_a$ & $V_b$ as in figure (a).

$$V_a = 2\Omega \times (-I_g) = -2V$$

$$V_b = 4\Omega \times (-I_9) = 4 \times -4 = -16V$$

$$\therefore V_{ab} = -2 - (-16V) = 14V$$

6a The equivalent resistance is found as shown in figures




(b) $V_a = V_2 = \frac{32 \times 141.93}{240 + 141.93} = 11.89V$

c) $V_1 = 32 - 11.89 = 20.108V$

d) $V_2 = 11.89V$

(e) As per fig (b).

$$I_1 = \frac{20.108}{400} = 0.05027 A$$

$$I_2 = 0.029725 A$$

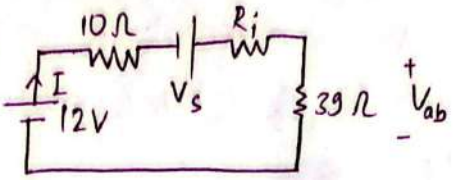
$$I = I_1 - I_2 = 0.02054A$$

The direction will be as marked in fig (b)

(7)(a) The circuit with voltage source becomes as shown below

Here, $V_s = 2A \times 6.8\Omega = 13.6V$

$R_i = 6.8\Omega$



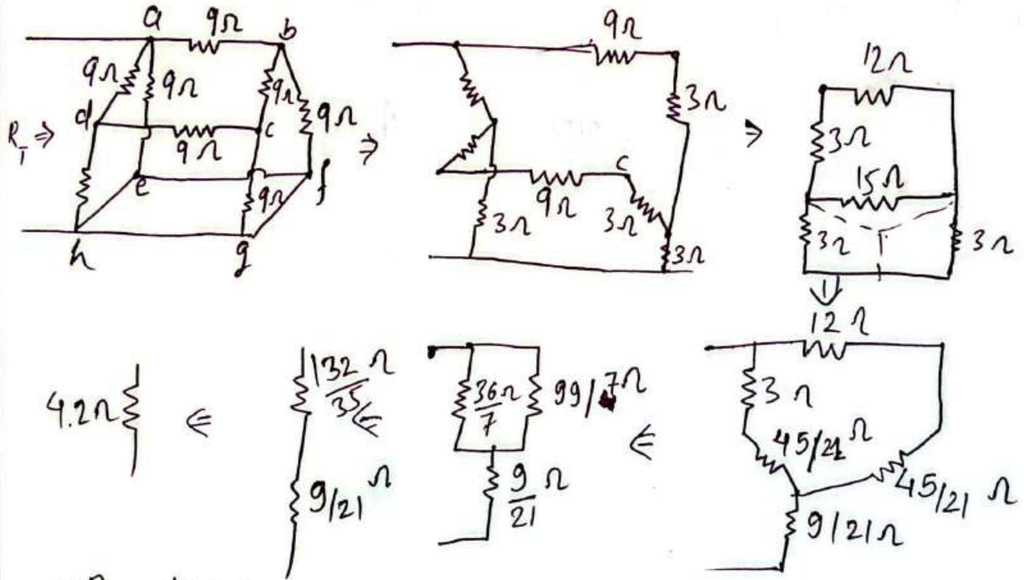

(b) Here,

The magnitude of current $(I) = \dfrac{(12+13.6)V}{(10+6.8+39)\Omega} = 0.45878A$

the direction would current coming out from the +ve terminal of the 12 V source as shown in figure above.

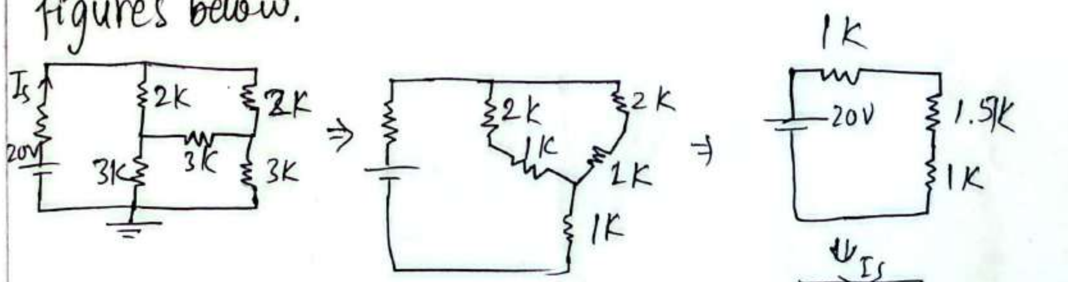
(c) The polarity of the voltage $V_{ab}$ is as marked in above figure, The magnitude of $V_{ab} = 39\Omega \times 0.45878 = 17.89V$

(8) The total resistance of the network is determined as shown in figure below:

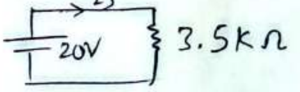



$\therefore R_T = 4.2\Omega$

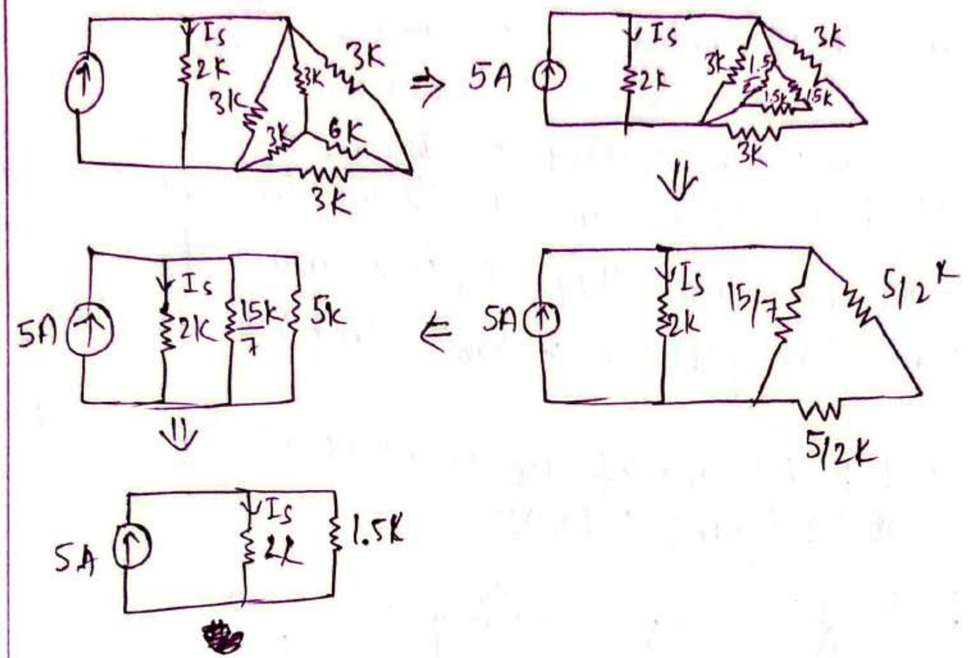
(9) To get the current $I_s$, the equivalent resistance across the voltage source (20V) is determined as shown in figures below.




$$I_s = \frac{20V}{3.5 \times 1000} = 5.714mA$$



10) To determine the current, the equivalent resistance for the Δ & Y network is ~~equa~~ evaluated as shown below & $I_s$ is determined using current divider rule.




$$\because\ I_s = \frac{5 \times 1.5 \times 1000}{2000 + 1500} = 2.14\ A$$